Experiment No. 8

Laplace Transform, Inverse Transform, ROC and Stability

OBJECTIVE:

- To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.
- To analyze system stability using poles, zeros, and the Region of Convergence (ROC) and verify the stability conditions of continuous-time systems.

I. To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.

Experimental Procedure:

- Define X(s) symbolically.
- Apply ilaplace to obtain x(t).
- Evaluate x(t) over $0 \leq t \leq 5$ s.
- Plot the reconstructed time-domain signal.
- Compare the plotted response with the analytical expression.

MATLAB Coding:

```
clc; clear; close all;
%% 8(ii) Determination of Inverse Laplace Transform

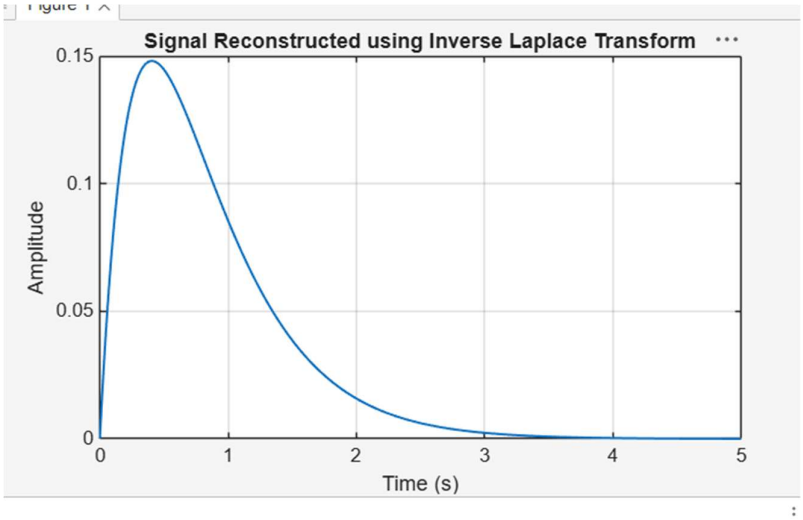
figure;
% Inverse Laplace Transform
% Requires Symbolic Math Toolbox
syms s t
X = 1/((s+2)*(s+3));
x = ilaplace(X,s,t);

disp('Inverse Laplace Transform x(t) =');
pretty(x);

tt = 0:0.001:5;
xx = double(subs(x,t,tt));

plot(tt,xx,'LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

Results:



II. To analyze system stability using poles, zeros, and the Region of Convergence (ROC) and verify the stability conditions of continuous-time systems.

Experimental Procedure:

- Define numerator and denominator coefficients.
- Calculate zeros and poles using roots.
- Generate the unit circle.
- Plot zeros and poles in the z-plane.
- Compare pole magnitudes with the unit-circle radius and report the causal-ROC stability result.

MATLAB Coding:

```
clc; clear; close all;
%% 8(iii) Analysis of Region of Convergence (ROC)

figure;
% ROC Analysis for a Rational Laplace Transform
% Requires Symbolic Math Toolbox
% X(s) = 1/((s+2)(s+4))
% Poles are at s = -2 and s = -4.

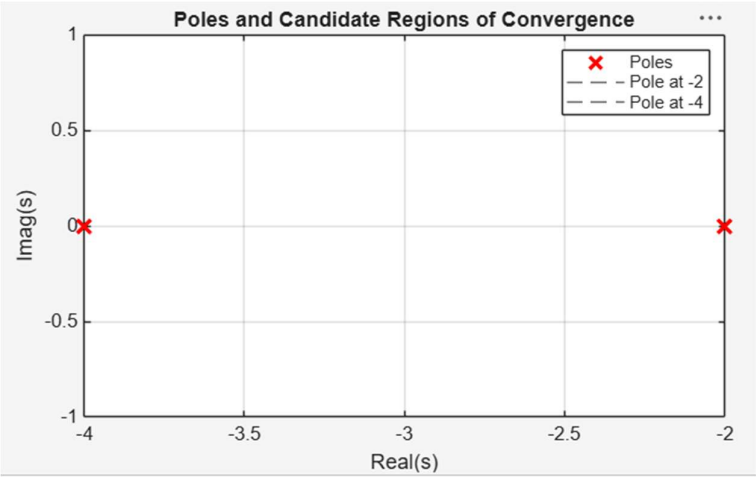
p = [-2 -4];

plot(real(p),imag(p),'rx','MarkerSize',10,'LineWidth',2);
hold on;
xline(-2,'--');
xline(-4,'--');
grid on;

xlabel('Real(s)');
ylabel('Imag(s)');
title('Poles and Candidate Regions of Convergence');
legend('Poles','Pole at -2','Pole at -4');

fprintf('Poles are at s = -2 and s = -4.\n');
fprintf('For a right-sided signal, ROC: Re(s) > -2.\n');
```

Results:


Poles and Candidate Regions of Convergence
Poles
Pole at -2
Pole at -4
1
0.5
0
-0.5
-1
Imag(s)
-4
-3.5
-3
-2.5
-2
Real(s)